

M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 10 (10TH MATRIC) - PHYSICS

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks (Theory: 60 | Practical: 15) | Time: 2 Hours 45 Minutes

SECTION 1: BISE GUJRWANWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- Section A: Objective (MCQs):

12 Compulsory MCQs (12 Marks).

- Section B: Short Questions:

Attempt 15 out of 24 Short Questions (Q2, Q3, Q4: 5/8 each = 30 Marks).

- Section C: Long / Numerical Questions:

Attempt 2 out of 3 Long Questions (Part a: Theory 5 Marks + Part b: Numerical 4 Marks = 18 Marks).

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Unit 10: Simple Harmonic Motion and Waves:

SHM conditions (a prop to -x), Mass-spring system, Simple Pendulum ($T = 2\pi\sqrt{L/g}$), Ball and bowl system, Wave motion, Mechanical vs Electromagnetic waves, Transverse vs Longitudinal waves, Wave equation ($v = f\lambda$), Ripple tank experiments (reflection, refraction, diffraction).

Unit 11: Sound:

Production and propagation of sound waves, Speed of sound in air ($v \sim 343$ m/s), Characteristics of sound (Loudness, Pitch, Quality/Timbre, Intensity), Decibel scale (Sound Level = $10 \log(I/I_0)$ dB), Echo, Audible frequency range (20 Hz to 20,000 Hz), Ultrasound applications in medicine and sonar.

Unit 12: Geometrical Optics:

Reflection of Light, Spherical Mirrors (Concave & Convex), Mirror Formula ($1/f = 1/p + 1/q$), Refraction of light, Snell Law ($n = \sin i / \sin r$), Total Internal Reflection (critical angle, optical fibers, periscope), Lenses (Convex & Concave), Lens formula, Defects of vision (Myopia vs Hypermetropia), Microscope and Telescope.

Unit 13: Electrostatics:

Production of electric charges, Electrostatic induction, Gold Leaf Electroscope, Coulomb Law ($F = k \cdot q_1 \cdot q_2 / r^2$, $k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$), Electric field and intensity ($E = F/q$), Electric potential ($V = W/q$), Capacitors ($C = Q/V$, Farad), Parallel and Series combinations of capacitors, Applications (photocopier, inkjet printer).

Unit 14: Current Electricity:

Electric current ($I = Q/t$, Ampere), Potential difference and EMF, Ohm Law ($V = IR$), Ohmic vs Non-Ohmic conductors, Factors affecting resistance ($R = \rho \cdot L/A$), Series and Parallel combination of resistors, Electric power ($P = VI = I^2R = V^2/R$), Kilowatt-hour ($1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$), Safe use of electricity (Fuse, Breaker, Earth Wire).

Unit 15: Electromagnetism:

Magnetic effect of steady current (Right Hand Rule), Force on current conductor in magnetic field ($F = ILB \sin \theta$), Fleming Left-Hand Rule, DC Motor, Electromagnetic Induction (Faraday and Lenz Laws), Mutual Induction, AC Generator, Transformer ($V_s/V_p = N_s/N_p$).

Unit 16: Basic Electronics:

Thermionic emission, Cathode Ray Oscilloscope (CRO components & uses), Analogue vs Digital electronics, Logic Gates: AND ($Y = A \cdot B$), OR ($Y = A + B$), NOT ($Y = A'$), NAND, NOR truth tables and symbols.

Unit 17: Information and Communication Technology (ICT):

Components of CBIS (Hardware, Software, Data, Procedures, People), Transmission of light signals via optical fibers, Radio waves, Satellite communication, Internet, Email, Browsers, Data storage devices (Hard disk, Flash drive).

Unit 18: Atomic and Nuclear Physics:

Atom and Nucleus (Protons, Neutrons, Nucleons), Isotopes, Natural Radioactivity, Alpha, Beta, Gamma radiation properties, Nuclear Decay, Half-Life ($N = N_0 (1/2)^n$), Nuclear Fission and Fusion reactions, Radiation hazards and safety.

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

- * In optics numericals, maintain sign convention strictly (concave mirror f is positive, convex f is negative; real images q is positive, virtual q is negative).
- * Draw neat circuit diagrams showing correct voltmeter (parallel) and ammeter (series) connections.